

Stress Test Two: A Two-Column Paper

C. Columns

Input: data $\{(x_i, y_i)\}_{i=1}^N$, rate η
for $t = 1$ **to** T **do**
 sample batch B ;
 $g \leftarrow \frac{1}{|B|} \sum_{i \in B} \nabla \ell_i$;
 $\theta \leftarrow \theta - \eta g$;
end
Output: θ

Algorithm 1: Training loop

Abstract

Two-column layouts are the norm in engineering venues. This paper mixes running text, equations that fill a narrow column, a figure spanning both columns, a wide table and an algorithm.

1 Introduction

Neural networks compute $y = \sigma(Wx + b)$ where σ is a nonlinearity such as the rectified linear unit $\text{ReLU}(z) = \max(0, z)$. Training minimises the empirical risk

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_\theta(x_i), y_i) + \lambda \|\theta\|_2^2, \quad (1)$$

with gradient descent $\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t)$. The learning rate η matters a great deal.¹ Results are summarised in Table 1 and Figure 1.

The softmax of a vector $z \in \mathbb{R}^K$ is

$$\text{softmax}(z)_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}. \quad (2)$$

Attention combines queries, keys and values:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V. \quad (3)$$

2 Method

We use the procedure in Algorithm 1. Each step samples a batch $B \subset \{1, \dots, N\}$ of size $|B| = 64$.

The model has three parts. First, an encoder maps the input to a hidden state. Second, a decoder reads the hidden state. Third, a small head predicts the label. Each part is trained jointly, and we report the average over five random seeds. This paragraph is deliberately long so that it flows from one column into the next one, which is where many converters lose the reading order or mix the two columns together line by line. A careful reader should see this sentence exactly once and in the right place.

¹See the appendix for the schedule we used.

3 Results

Our model improves accuracy by 2.8 points while using fewer parameters than *Deep*. The perplexity drops from 16.9 to 15.8, a relative gain of $\approx 6.5\%$. Error rates fall as well.

4 Conclusion

We presented a stress test. The inequality $\mathbb{E}[\|g\|^2] \leq G^2$ bounds the gradient norm, and the bound $\mathcal{O}(1/\sqrt{T})$ follows. Future work includes k -fold validation with $k \in \{5, 10\}$.

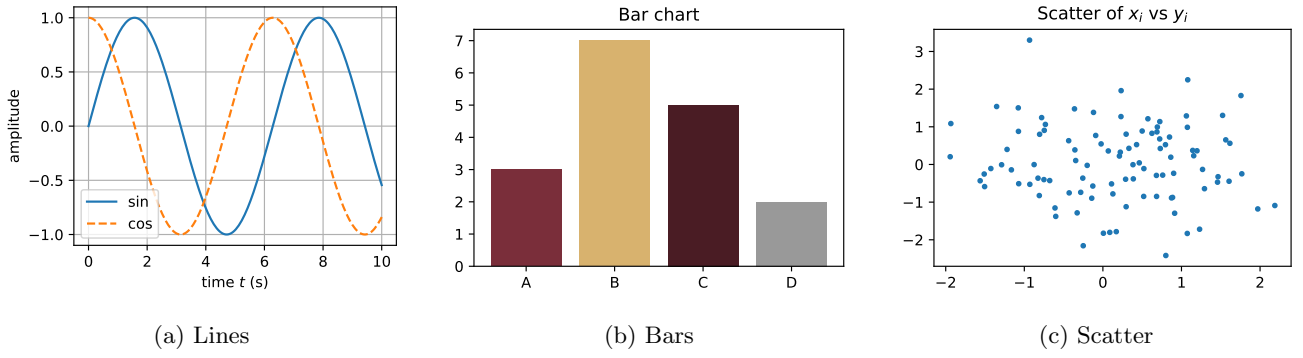


Figure 1: Three panels side by side spanning both columns. Panel (a) shows periodic signals, (b) a bar chart and (c) a scatter plot.

Table 1: Main results on six benchmarks. Best numbers are in bold.

Method	Vision		Language		Audio		Avg.	Params
	Top-1	Top-5	BLEU	PPL	WER	CER		
Baseline	71.2	90.1	27.3	18.4	9.1	3.2	36.6	25M
Wide	73.5	91.0	28.1	17.2	8.7	3.0	37.6	60M
Deep	74.0	91.8	29.4	16.9	8.5	2.9	38.3	80M
Ours	76.8	93.2	29.1	15.8	7.9	2.6	39.2	45M

Table 2: Ablation with grid lines.

Variant	Acc.	Δ
Full model	76.8	—
no attention	74.1	−2.7
no dropout	75.9	−0.9